

Module 01 | Introduction to Deep Learning and Perceptrons

My Solution to Practice Problems

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Bias

What is a bias in a perceptron?

The Bias (b) is a parameter of a Perceptron, additional to Input Weights ($W = [w_i]$), that the perceptron learns during training, that adjusts the Weighted Sum ($W \cdot X$) to the threshold of the Activation Function that follows.

Why do we add bias to a perceptron?

We add Bias (b) to the Weighted Sum ($W \cdot X$) to adjust the weighted sum to the threshold of the Activation Function of the perceptron.

For `SKLearn#Perceptron`, the Activation Function is the Step Function as defined below.

$$y_{step}(\text{Weighted Sum}) = \begin{cases} 1, & \text{Weighted Sum} \geq 0 \\ 0, & \text{Weighted Sum} < 0 \end{cases}$$

This 0 is the (Decision) Threshold Of the Activation Function, for `SKLearn#Perceptron`.

During training, `SKLearn#Perceptron` finds optimal b so that adding it to the weighted sum adjusts the sum to the threshold 0, for the sake of correct activation decision.

Weights

What are weights in a perceptron?

Weights ($W = [w_i]$) are the measures of importance of the individual features/inputs toward activation of the perceptron.

How do weights affect the output of a perceptron?

Suppose, from training, the Perceptron learns that the weight of input x_i is w_i .

Influence of a Feature on Activation Decision: $\begin{cases} \text{None,} & \text{if } w_i \text{ close to } 0 \\ \text{Strongly Direct,} & \text{if } w_i \text{ largely positive} \\ \text{Strongly Inverse,} & \text{if } w_i \text{ largely negative} \end{cases}$

Perceptron Output

A perceptron has inputs $x_1 = 2$, $x_2 = 3$, weights $w_1 = 0.5$, $w_2 = 1$, and bias $b = 1$. Calculate the weighted sum (before activation).

We have,

$$\begin{aligned} X &= [x_1 \quad x_2] \\ &= [2 \quad 3] \end{aligned}$$

$$\begin{aligned} W &= [w_1 \quad w_2] \\ &= [0.5 \quad 1] \end{aligned}$$

$$b = 1$$

$$\begin{aligned} W \cdot X + b &= [0.5 \quad 1] \cdot [2 \quad 3] + 1 \\ &= 1 + 3 + 1 \\ &= 5 \end{aligned}$$

So, the weighted sum is 5.

Step function

What is the step function and how does it work?

Step functions, as defined below, are one kind of activation functions used in a perceptron.

$$y_{step}(\text{Weighted Sum}) = \begin{cases} 1, & \text{Weighted Sum} \geq \text{Threshold} \\ 0, & \text{Weighted Sum} < \text{Threshold} \end{cases}$$

For `SKLearn#Perceptron`, this `Threshold = 0`.

The `Weighted Sum` is the input to a step function, as to any other activation function. As obvious from the definition above, if the `Weighted Sum` is greater than or equal to the `Threshold` value, the perceptron (or the artificial neuron) fires, or gets activated, that is the output of the activation function is `1`.

Otherwise, it is `0` — the neuron does not fire— it remains inactive.

Activation Function

Why do we need an activation function in a perceptron?

To decide if the input values are, collectively, `important enough` for the perceptron to fire.

The `importance` is represented by the `weighted Sum`, which is the input to the activation function. And the activation function decides, based on `Threshold` inside the function, if the importance is `enough`.

Name three common activation functions.

1. Sigmoid
2. tanh
3. ReLU